
LECTURE 3: MR. SOLOMON

VENOUS DRAINAGE OF LOWER LIMB

1. WHERE ARE SUPERFICIAL STRUCTURES?

Superficial structures are present in the **subcutaneous (adipose) tissue**, just below the skin.

These include:

- **Superficial veins**
- **Sensory (cutaneous) nerves**
- Superficial lymphatics

Important rule (very high-yield):

- **Sensory nerves → superficial**
- **Motor nerves → deep**

Why?

Sensory nerves supply skin → they must reach the skin.

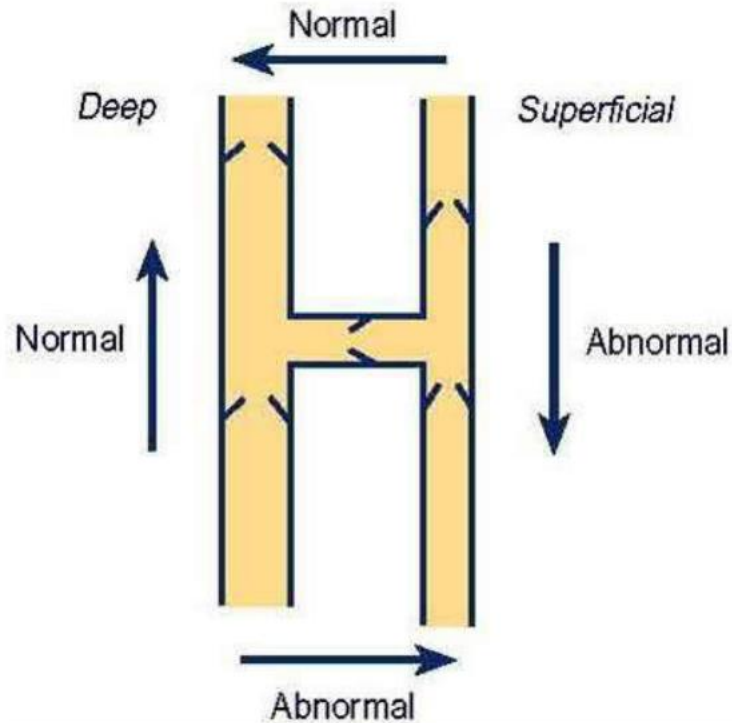
Motor nerves supply muscles → muscles lie deep.

2. DIRECTION OF VENOUS BLOOD FLOW

Under **normal conditions**, venous blood flows:

1. **From inferior to superior**
(from foot → leg → thigh → heart)
2. **From superficial veins to deep veins**

This is the DESIGNED pathway.



3. ABNORMAL VENOUS FLOW

Abnormal flow is the **reverse**:

- From **deep** → **superficial**
- Or downward due to gravity

This happens when **venous valves become incompetent**, leading to **varicose veins**

4. WHY DOES BLOOD FLOW FROM SUPERFICIAL → DEEP VEINS?

This is the most important concept.

SUPERFICIAL VEINS

They are **not meant to fight gravity**.

Characteristics:

- Thinner wall
- Less smooth muscle
- Poor external support (lying in fat)
- Narrow **lumen**
- Less efficient valves

So they are **collection channels**, not pumping channels.

DEEP VEINS

They are **designed for upward transport**.

Characteristics:

- Thicker wall
- More smooth muscle
- Firmly supported by surrounding muscles
- Run **alongside arteries**
- Strong valves

5. TWO MAJOR FORCES PUSHING BLOOD UPWARD IN DEEP VEINS

1. MUSCLE PUMP

- When leg muscles contract, they squeeze deep veins.
- Blood is forced upward.
- Valves prevent backward flow.

This is why walking helps venous return.

2. ARTERIAL PULSATIONS

- Deep veins lie next to arteries.
- Each arterial pulse compresses the vein.
- This helps push blood **against gravity**.

SUMMARY

Blood flows from **superficial veins to deep veins** because:

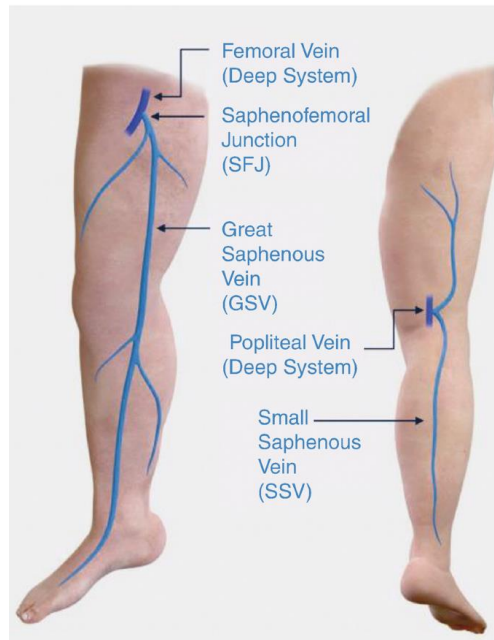
- Superficial veins are thin-walled and poorly supported
- Deep veins are thick-walled and surrounded by muscles
- Muscle contraction and arterial pulsation help propel blood upward
- Venous valves ensure one-way flow

VEINS OF THE LOWER LIMB: SUPERFICIAL AND DEEP SYSTEMS

The veins of the lower limb are divided into **superficial** (located in the subcutaneous tissue) and **deep** (located beneath the deep fascia, accompanying major arteries). Perforating (or perforator) veins connect the two systems, allowing blood to flow from superficial to deep veins. Valves in these veins ensure unidirectional flow toward the heart, and muscle contractions (especially in the calf) act as a "peripheral heart" to pump blood upward against gravity.

SUPERFICIAL VEINS

These are the **great saphenous vein (GSV)** and **small saphenous vein (SSV)**. They drain blood from the skin and superficial tissues.



- **Great Saphenous Vein (GSV):**
 - Origin: Union of the **dorsal vein of the great toe** and the **dorsal venous arch on the dorsum of the foot**.
 - Course: Ascends anterior to the medial malleolus, along the medial leg and thigh, and enters the femoral vein at the saphenofemoral junction (in the groin, through the saphenous hiatus in the fascia lata).
 - Key features: Longest vein in the body; often visible ("saphenous" means visible). Runs anterior to the medial malleolus; accompanied Anteriorly by the saphenous nerve (a sensory branch of the femoral nerve) in the lower leg — be cautious during procedures to avoid nerve injury.
 - Clinical note: Commonly involved in varicose veins; used as a graft in coronary artery bypass surgery.

- **Small Saphenous Vein (SSV):**
 - Origin: **Dorsal vein of the little toe and dorsal venous arch.**
 - Course: Ascends posterior to the lateral malleolus, along the posterior calf (lateral to the calcaneal tendon), and drains into the popliteal vein in the popliteal fossa (may pierce the deep fascia).
 - Key features: Drains the lateral foot and posterior leg; often accompanied by the sural nerve.

DEEP VEINS

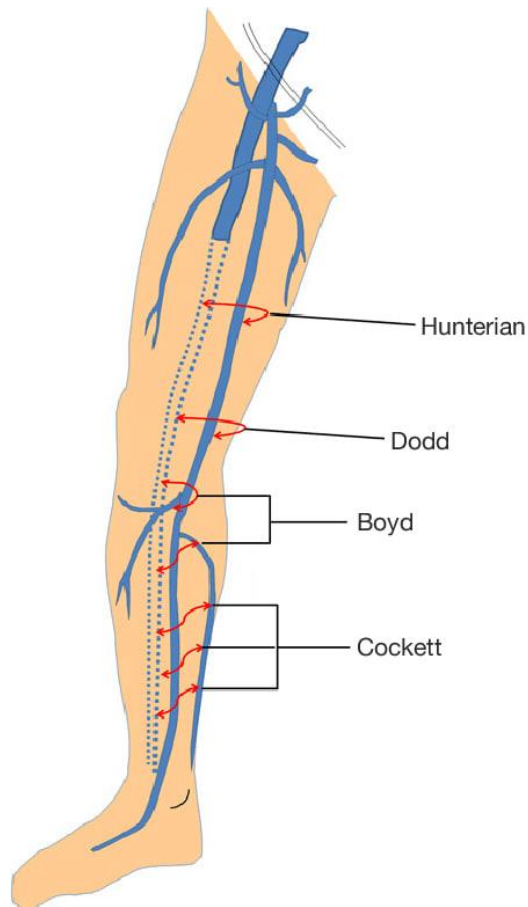
These accompany the major arteries and are crucial for returning blood from muscles and deeper tissues.

- **Popliteal vein:** Formed by the union of anterior and posterior tibial veins; drains into the femoral vein.
- **Femoral vein:** Continues from the popliteal vein; ascends through the thigh and becomes the external iliac vein.
- **Profunda femoris vein:** Accompanies the profunda femoris artery; drains the thigh muscles.
- **Anterior and posterior tibial veins:** Drain the anterior and posterior compartments of the leg, respectively.

PERFORATING VEINS (PERFORATORS)

These connect superficial to deep veins. Incompetent perforators can contribute to varicose veins and venous stasis ulcers. Key named perforators (locations to remember):

- **Hunter's perforator:** Mid-thigh (adductor canal region).
- **Dodd's perforator:** In the adductor canal (thigh).
- **Boyd's perforator:** Below the knee (paratibial).
- **Cockett's perforators:** Mid-calf and ankle (posterior tibial; superior, middle, and inferior).
- **Lateral ankle perforators:** Around the lateral malleolus.



ANATOMICAL NOTES ON PROCEDURES INVOLVING GSV/SSV

- **Saphenous nerve:** Runs anterior to the GSV in the lower leg; close contact increases risk of injury during vein harvesting or stripping (causing numbness or neuralgia).
- **Anesthesia for GSV procedures:** Local anesthesia or saphenous nerve block (sub-sartorial) is used to numb the nerve. Locate near the patella or in the adductor canal (deep to sartorius muscle).
- **Surgical techniques:** Transillumination improves vein visibility; incisions avoid nerve damage.

PERIPHERAL HEART: CALF MUSCLE PUMP

The **soleus muscle** (with gastrocnemius) forms the calf pump, often called the "peripheral heart." The soleus contains venous sinuses filled with blood.

Contraction pumps blood upward (into popliteal → femoral → external iliac → common iliac veins); relaxation allows refilling. Dysfunction leads to venous stasis and varicose veins.

VARICOSE VEINS

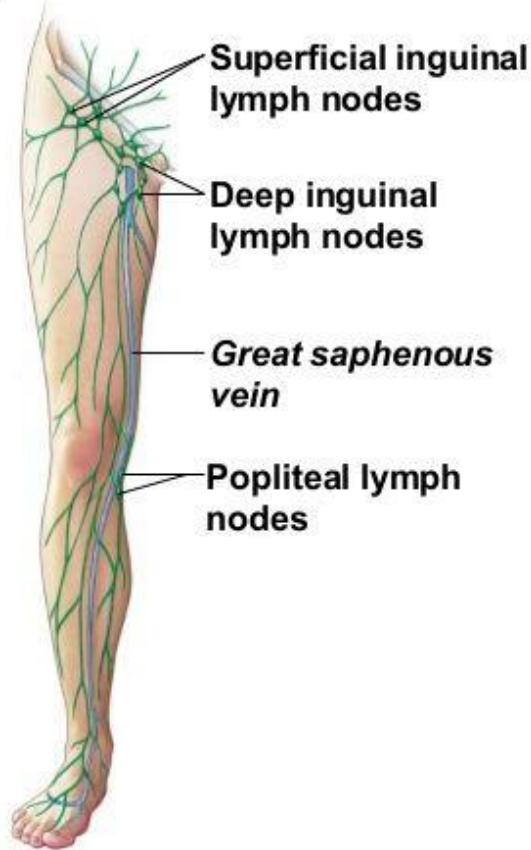
Varicose veins are dilated, tortuous superficial veins due to valve incompetence, wall weakness, or increased venous pressure.

- **Common causes/risk factors:**
 - Prolonged standing/sitting (e.g., occupations like nursing).
 - Pregnancy (hormonal changes, increased abdominal pressure).
 - Obesity, family history, aging.
 - Deep vein issues (e.g., post-thrombotic syndrome).
 - Femoral vein cannulation (prolonged use blocks venous return, leading to superficial vein dilation).
- **Complications:** Pain, swelling, ulcers, thrombosis, bleeding.

LYMPHATIC DRAINAGE OF THE LOWER LIMB

Lymphatics follow veins and are divided into superficial and deep systems.

Figure 23.12 Lymphatic Drainage of the Lower Limb



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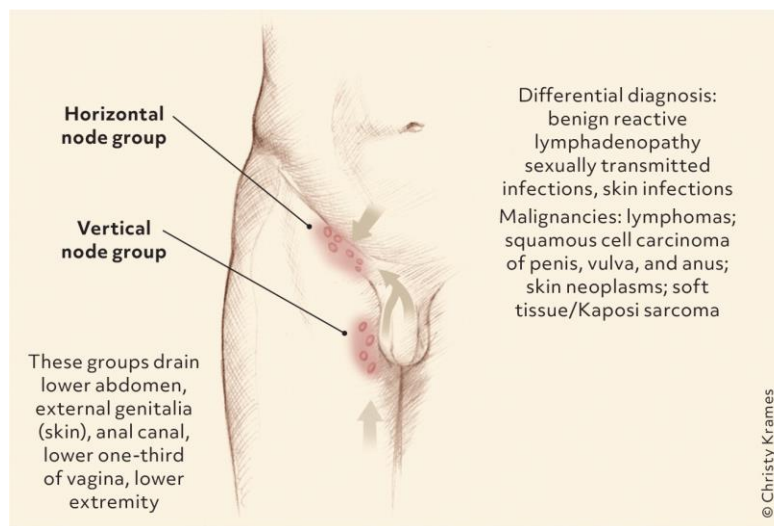
- **Superficial lymphatics:** Accompany superficial veins (GSV and SSV). Drain the skin and subcutaneous tissue.
 - Medial (with GSV): To superficial inguinal lymph nodes.
 - Lateral/posterior (with SSV): To superficial popliteal lymph nodes → deep popliteal or inguinal nodes.
- **Deep lymphatics:** Accompany deep veins; drain muscles and joints → deep inguinal nodes.
- **Key nodes:**
 - Superficial popliteal: Drain lateral foot and posterior leg.
 - Deep popliteal: Drain deeper structures of leg.
 - Inguinal (superficial and deep): Final drainage to iliac nodes.

1. INGUINAL LYMPHADENOPATHY AND GROIN SWELLING

Inflamed or painful inguinal lymph nodes (LNs) indicate pathology in their drainage area, which includes the lower limb, perineum, gluteal region, and external genitalia (e.g., scrotum in males). Always examine these areas thoroughly—not just the leg.

- **Lymphatic drainage pathway:**

- Gluteal region → Superficial and deep inguinal LNs.
- Scrotum (and penis, vulva, perineum) → Superficial inguinal LNs.

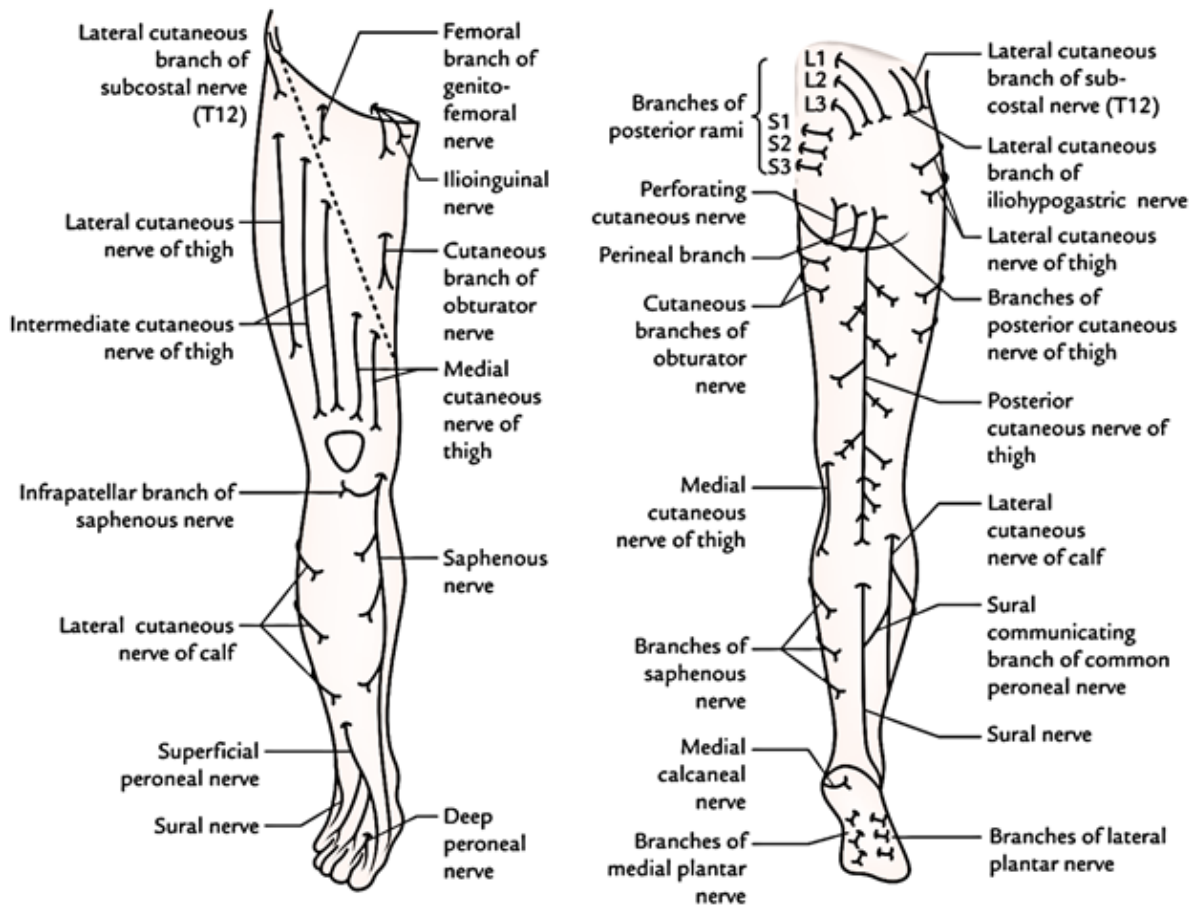


Differential diagnosis for groin swelling / inguinal lymphadenopathy:

- Inguinal lymphadenopathy (infection, malignancy, etc.).
- Dilated great saphenous vein (saphenous varix/varicose vein).
- Benign tumor (lipoma).
- Femoral hernia.
- Local infection (boil/abscess in drainage area).
- Filariasis (parasitic infection causing elephantiasis).
- Hodgkin's lymphoma (or other malignancies).

2. CUTANEOUS NERVES OF THE LOWER LIMB

Cutaneous nerves provide sensation to the skin and are derived from two main plexuses.



- **Lumbar Plexus** (L1–L4 roots): Supplies anterior abdominal wall and lower limb.

Key branches:

- Femoral nerve,
- obturator nerve,
- lateral cutaneous nerve of thigh,
- genitofemoral nerve,
- saphenous nerve (terminal branch of femoral nerve).

- **Sacral Plexus** (L4–S4 roots): Supplies posterior thigh, leg, and perineum.
 - Sciatic nerve (L4–S3).
 - Pudendal nerve (S2–S4).
 - Superior gluteal nerve (L4–S1).
 - Inferior gluteal nerve (L5–S2).

Note: The inferior gluteal nerve is separate from the sciatic nerve; it innervates gluteus maximus only.

- **Clinical relevance:**
 - Posterior leg and popliteal fossa sensation → Primarily S2 dermatome (sudden sensory loss here may indicate sciatic nerve or S2 root pathology).
 - Anesthesia in femoral triangle (e.g., for incisions): Block femoral nerve and genitofemoral nerve (femoral branch) above the inguinal ligament.

3. DERMATOMES OF THE LOWER LIMB

Dermatomes are skin areas supplied by a single spinal nerve root. There is some overlap between adjacent dermatomes.

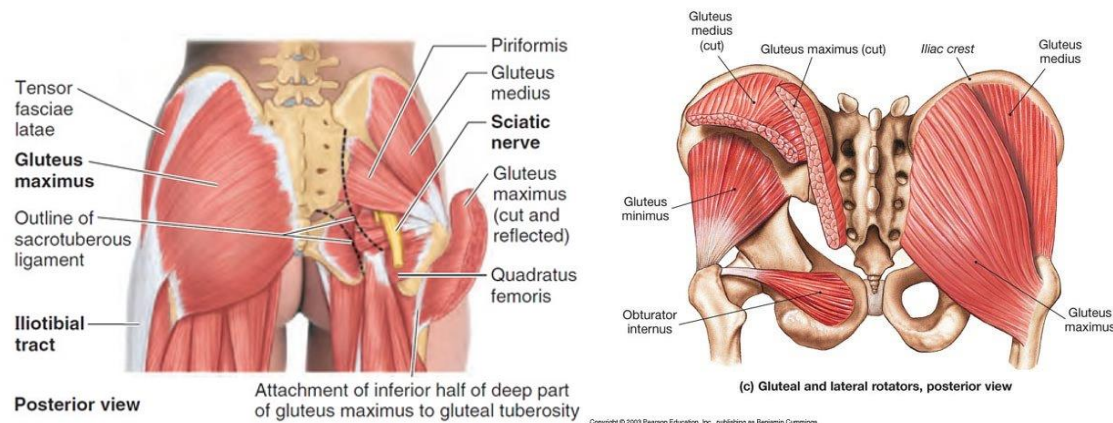
Key dermatomes:

- L1–L3: Anterior thigh.
- L4: Medial leg and medial foot.
- L5: Lateral leg, dorsum of foot.
- S1: Lateral foot, sole.
- S2: Posterior thigh and leg (popliteal fossa).

4. DEEP GLUTEAL MUSCLES (POSTERIOR COMPARTMENT OF THIGH)

These muscles are lateral rotators of the hip and lie deep in the gluteal region.

Mnemonic: **PGOGQ** (Piriformis, Gemellus superior, Obturator internus, Gemellus inferior, Quadratus femoris).



- **Innervation:**
 - Piriformis → Nerve to piriformis.
 - Obturator internus + Gemellus superior → Nerve to obturator internus.
 - Quadratus femoris + Gemellus inferior → Nerve to quadratus femoris.
- All are lateral (external) rotators of the hip.

5. TRENDELENBURG SIGN AND GAIT (HIP STABILITY)

START WITH THE normal SITUATION (THIS IS THE REFERENCE FRAME)

When a person **stands on one leg**, three things are happening:



1. The entire **body weight** acts downward on the pelvis.
2. The pelvis WANTS to tilt and drop on the side of the lifted leg.
3. Something must prevent that drop.

That “something” is the **hip abductors of the stance limb**:

- **Gluteus medius**
- **Gluteus minimus**
- (assisted by tensor fasciae latae)

These muscles pull the pelvis **upward on the opposite side**, keeping it level.

EXAMPLE 1: YOU LIFT THE LEFT LEG

- **RIGHT leg** is on the ground (stance limb)
- **RIGHT gluteus medius & minimus contract**
- They pull the **left side of the pelvis upward**
- Result: **Pelvis stays level**
- Trunk remains upright

EXAMPLE 2: YOU LIFT THE RIGHT LEG

- **LEFT leg** is on the ground
- **LEFT gluteus medius & minimus contract**
- They pull the **right side of the pelvis upward**
- Result: **Pelvis stays level**

➡ Still normal

Key normal rule:

When you stand on one leg, the pelvis should remain level(Straight).

APPLY THIS TO NORMAL WALKING

Walking is basically **controlled single-leg standing**.

- Right leg on the ground → left leg is in the air
- Left leg on the ground → right leg is in the air

In each step:

- The **stance-side abductors contract**
- The pelvis stays horizontal
- The trunk stays upright

This is **normal gait**.

WHAT GOES WRONG IN TRENDELENBURG SIGN?

Now imagine the **hip abductors are weak or paralyzed** on one side.

Causes commonly include:

- Injury to **superior gluteal nerve**
- Gluteus medius/minimus weakness
- Hip joint disease, fracture, congenital dislocation

(let's assume **RIGHT hip abductor weakness**)

EXAMPLE : YOU LIFT THE LEFT LEG (PROBLEM APPEARS HERE)

- **RIGHT leg** is on the ground
- **RIGHT gluteus medius/minimus are weak**
- They **cannot hold the pelvis level**
- The **LEFT side of the pelvis DROPS**

✦ **This is a POSITIVE Trendelenburg sign**

Important exam rule:

Pelvis drops on the **side of the lifted leg**

Weakness is on the **standing leg!**

So here:

- Drop → **LEFT**
- Weakness → **RIGHT**

EXAMPLE : YOU LIFT THE RIGHT LEG

- **LEFT leg** is on the ground
- **LEFT abductors are normal**
- Pelvis stays level

➡ No problem on this side

NOW TURN THIS INTO WALKING (TRENDLENBURG GAIT)

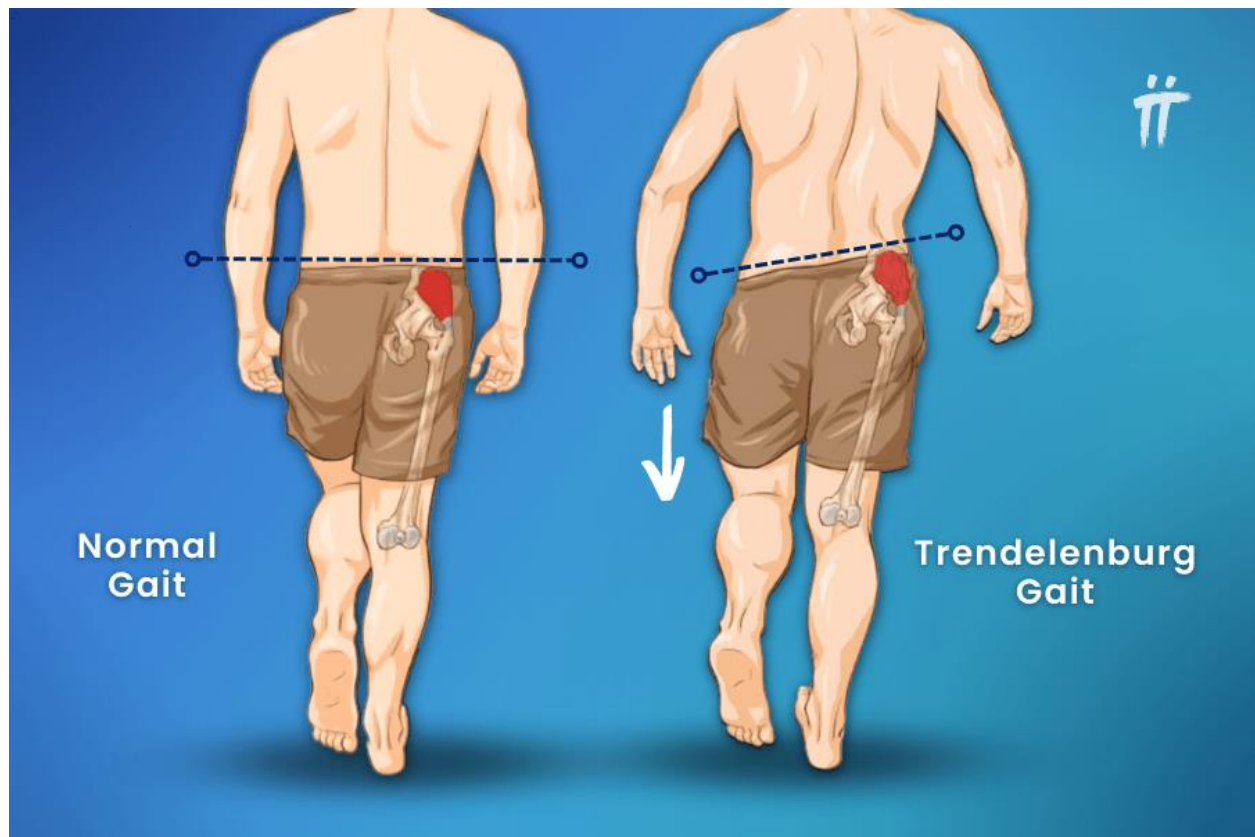
STEP 1: PERSON STEPS FORWARD WITH THE LEFT LEG

- **RIGHT leg** is supporting body weight
- **RIGHT abductors are weak**
- Pelvis starts to **drop on the LEFT side**

STEP 2: BODY COMPENSATES

- Person **leans trunk to the RIGHT side**
- This reduces body-weight pull
- Makes walking possible despite weakness

➡ This sideways trunk lean is **Trendelenburg gait**



SUMMARY

- **Lift LEFT → pelvis drops LEFT → problem is RIGHT**
- **Lift RIGHT → pelvis drops RIGHT → problem is LEFT**

END OF LECTURE 3:

Read on Muscles and Innervation by yourself since he didn't teach thoroughly!

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